

Polymers and plastics

COMPOUNDS FORMED FROM LONG CHAINS OF SMALLER MOLECULES ARE CALLED POLYMERS.

SEE ALSO

◀ 84–85 Genetics

◀ 96–97 Properties of materials

◀ 158–159 Hydrocarbons

Stretching and deforming

174–175 ▶

Plastics and other artificial fibers, such as nylon, are familiar types of polymers. However, these long-chained molecules are also widespread in nature. Many of the chemicals in food are polymers too.

Monomers

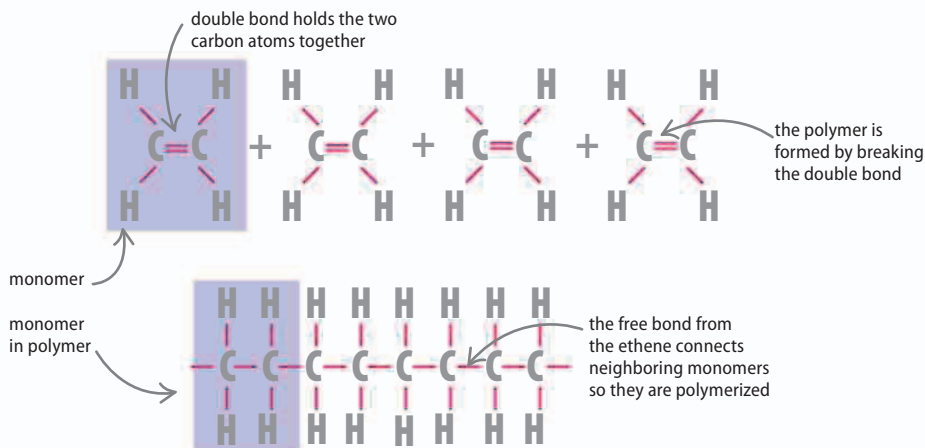
The repeating units in a polymer are called monomers. A polymer may contain a single type of monomer or have two or more types of repeating units—known as copolymers. Monomers are held together by covalent bonds (see page 114). Many artificial polymers are derived from alkene monomers, which have double bonds that can be broken and reformed to make the chains.

▷ Ethene monomer

One of the most common plastics is made from chains of ethene monomers. Ethene is the simplest alkene molecule. Its polymer is called polythene.

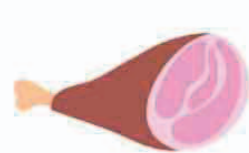
▷ Polythene polymer

While ethene is a gas, polythene (also known as polyethylene) is a transparent solid. It can be formed from an unlimited number of ethene monomers.



Natural polymers

The natural world contains many polymers. Living things frequently digest these large compounds, breaking them into their monomers, which are absorbed and then rebuilt into different polymers.



△ Protein

Muscles and many other features in a living body are made from proteins, which are polymers of amino acid monomers.



△ DNA

DNA is a complex copolymer. The sides are formed from chains of sugar, while the crosslinks are pairs of four monomers called nucleic acids.



△ Cellulose

The wall around a plant cell is made from a polymer of glucose called cellulose. Cellulose forms tough fiber and is a major component of wood and paper.



△ Starch

Found in potatoes and bread, starch is also made from glucose monomers. However, they are chained together differently to form globules rather than fibers.